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**100-ft Ring Road, Bengaluru – 560 085, Karnataka, India**

***Capstone Project Report***
***on***

# Careers Page : A smartrecruit

*Submitted by*

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**Aug 2025 – Jan 2026**

**under the guidance of**

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# CERTIFICATE

*This is to certify that the project entitled.*

## Careers Page : A smartrecruit

*is a bonafide work carried out by*

## Gagan Naik - (PES1PG24CA357)

in partial fulfillment for the completion of Capstone Project Phase-2 work in the Program of Study MCA under rules and regulations of PES University, Bengaluru during the period Feb 2026 – May 2026. The project report has been approved as it

satisfies the academic requirements of 4th semester MCA.

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# DECLARATION

I, **Gagan Naik,** bearing **PESPG24CA152** hereby declare that the Capstone project phase-1 entitled, ***Careers Page:A smartrecruit,*** is an original work done by me under the guidance of **Mr. Santosh Katti,** Assistant Professor PES University, and is being submitted in partial fulfillment of the requirements for completion of 3$^{rd}$ Semester course in the Program of Study **MCA**. All corrections/suggestions indicated for internal assessment have been incorporated in the report.

The plagiarism check has been done for the report and is below the given threshold.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other course.

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# ABSTRACT

Recruitment in modern organizations faces challenges such as high application volume, manual resume screening, fragmented hiring tools, lack of transparency, and malpractice in online assessments. Traditional Applicant Tracking Systems (ATS) rely heavily on keyword-based filtering, leading to inaccurate candidate evaluation and biased outcomes. To address these limitations, this project proposes “Careers Page: A Smart Recruit”, an AI-powered recruitment automation platform that centralizes and optimizes the end-to-end hiring process.

The system integrates ATS-based resume scoring, online assessments with real-time proctoring, automated shortlisting, and recruiter analytics dashboards into a unified platform. Natural Language Processing (NLP) techniques such as Jaccard Similarity and Cosine Similarity are used to analyze resumes against job descriptions, extract skills, and evaluate experience relevance. A secure proctoring mechanism using face detection, tab-switch monitoring, and multiple-face alerts ensures assessment integrity.

The platform supports both candidates and recruiters through role-based dashboards, enabling efficient hiring decisions and improved candidate experience. Experimental analysis shows a significant reduction in screening time and improved selection accuracy. With its integration of AI, Machine Learning, Computer Vision, and Full-Stack development, the system demonstrates strong academic and real-world relevance, justifying its scope as a major capstone project.

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# Chapter 1

# INTRODUCTION

## 1.1 Project Description

The Careers Page of the Smart Recruit platform is an interactive and centralized platform that helps to bridge the gap between the candidate and the recruiter by providing a simplified recruitment process based on smart automation and unique assessment systems. It is intended to offer candidates an uncomplicated experience to browse and apply to vacancies, offer job seekers to submit their resumes, and take role-specific tests, and also allow recruiters to efficiently handle applications with AI-based analytics.

With the help of this module, applicants will be able to post their resumes in standardized formats, change personal and professional information, and take tests during the evaluation of skills depending on the situation and the job position. The system automatically parses the resumes that are submitted with intelligent parsing to retrieve the relevant information e.g. educational qualification, technical skills, certifications and experience levels. The AI engine uses these extracted features to create structured candidate profiles to be analyzed further.

On the recruiter side, the careers page offers a set of tools to browse through submissions of candidates, compare their performance under various evaluation criteria and get recommendations on suitability based on AI-driven insights developed using pre-defined job requirements. The built in evaluation system guarantees objective measurement of applicant competencies, thus eliminating manual screening, and subjective bias at the preliminary levels of the hiring process.

Moreover, the platform provides comparative analysis of the candidates through the introduction of structured performance indicators, alignment scores on skills, and indicators of recommendation. This would help the recruiters to make quicker and more informed hiring choices and at the same time be fair and consistent in the hiring process.

In general, the careers page is an essential part of the Smart Recruit system as it provides more transparency, greater accuracy of candidate-job matching, and higher overall efficiency of the recruitment lifecycle with intelligent automation and data-driven decision support.

## 1.2 Problem Definition

Traditional recruiting methods are based on manual resumes processing, email correspondence, spreadsheets, and various unintegrated applicant recruitment systems that considerably diminish the effectiveness and dependability of the candidate assessment. By quickly going through large numbers of resumes, recruiters tend to miss out on important details that do not have standardized evaluation criteria and thus; shortlisting of appropriate candidates is delayed. Moreover, this lack of formal automation makes it more likely to overlook qualified candidates and instill unwanted bias in the screening procedure. Candidates, in turn, are often exposed to low levels of transparency when it comes to parameters of evaluation, selection and updates on their application status, which influences their participation and trust in the recruitment process.

Moreover, most of the current online assessment systems do not incorporate proper monitoring systems to promote authenticity and fairness in evaluation of the candidates, which heightens chances of malpractice and unreliable test scores. The majority of the recruitment solutions on the market today only cover a single step of the hiring cycle e.g. resume filtering, assessment conduction or candidate tracking but not an end to end system. The result of this disjointed strategy is lack of coordination between recruitment stages, overlapping efforts and erratic decision making processes. Consequently, the necessity to have a single AI-enabled recruitment system that can guarantee the organization of workflow, transparent assessment systems, and credible assessment of candidates to improve the overall recruitment performance and the level of fairness is high.

## 1.3 Proposed Solution

The suggested system will present a smart careers page, which will facilitate the entire recruitment process, combining several AI-based assessment elements within one platform. It allows candidates to send resumes that are automatically evaluated by Applicant Tracking System (ATS)-based similarity scoring algorithms to quantify a fit to job descriptions. Also, the system can use Natural Language Processing (NLP) to assess the skills, experience levels, and relevance to the domain of the candidate more precisely than the traditional approach to filtering with the help of keywords. To enhance the assessment of candidates further, the platform will include online assessments in the form of MCQs and online coding assessments depending on the specific job position, so both the theoretical and practical problem-solving skills will be assessed. Proctoring mechanisms, which are real-time, improve the credibility of online tests by observing the activity of candidates during the evaluation process, which facilitates the overall fairness and authenticity.

In addition to the assessment of candidates, the platform also offers recruiters interactive dashboards that can display in-depth analytical data about the performance aspects of candidates, distribution of skills, and trends of hiring across various positions. Such dashboards allow recruiters to conduct comparative analysis, determine the best performing candidates more effectively and make decisions on shortlisting based on data more confidently and consistently. The system also promotes automated ranking schemes where applicants are ranked with respect to various assessment parameters yet enabling recruiters to have a final say and control over the selection process. In general, the suggested solution will improve the job posting process, efficiency of the workflow, and promote unbiased decision-making through intelligent automation and human knowledge instead of trying to substitute it entirely.

## 1.4 Purpose

This project aims to create and deploy a smart and trusted system of candidate recruitment support that would increase the efficiency of the recruitment process by AI-aided screening of candidates, but will not replace human control in decision-making. The idea behind the system is to substantially enhance resume to job matching accuracy, by processing the candidate skills, experience, educational background and qualification with advanced Natural Language Processing (NLP) techniques. The platform allows enhancing semantic interpretation of candidate profiles and job requirements by going beyond the conventional approach of filtering by key words, thus minimizing bias, enhancing the consistency of the screening process, and reducing the number of applicants-employer mismatches in the first phases of the recruitment process. Moreover, the system will be designed to assist the recruiters with information-driven insights that will enable them to make more objective and structured judgments without undermining their authority in the ultimate decision-making.

The other significant aim of the project is to have the integrity, transparency and fairness of the candidate assessments through the provision of secure online MCQ-based tests and coding evaluations with support of real time proctoring systems. These features can be used to uphold authenticity when remote assessment is being conducted, and minimize the possibility of malpractice, which enhances the credibility of evaluation results. Moreover, the platform is dedicated to improving the overall candidate experience with the help of transparent application processes, systematic assessment phases, and better insights into the hiring procedure. The system also saves a lot on the workload and time wastage on the recruiter by automating repetitive and time consuming screening processes like resume parsing, candidate ranking and performance analysis. Consequently, recruiters will have additional time to make strategic hiring choices, engage with candidates, and develop talent within the organization, which will eventually lead to a more efficient, scalable, and reliable recruitment ecosystem.

## 1.5 Scope

The system can be used by companies, colleges, startups, and HR consultancies for recruitment and screening activities. The scope includes resume analysis, assessments, proctoring, and analytical dashboards. Advanced interview intelligence and large-scale deployment are considered future enhancements.

# Chapter 2

# LITERATURE SURVEY

## 2.1 Domain Survey

Artificial Intelligence has increasingly been adopted in recruitment to assist resume screening, candidate evaluation, and hiring decisions. Research highlights the use of NLP-based similarity models, ethical AI practices, and computer vision techniques to improve fairness, transparency, and assessment integrity in recruitment systems.

## 2.2 Related Work

This includes reviewing research papers and journals, with the literature survey conducted based on the following studies.

- **The Use of Artificial Intelligence (AI) in the Hiring Process (2025)**

  Author: Hosain, M. S.

  Publication: ScienceDirect.

  Summary: Hosain investigates the role of AI in modern recruitment by analyzing applicant perceptions related to trust, fairness, and transparency. The study highlights that AI-driven systems significantly improve efficiency and reduce manual bias during resume screening and initial evaluations. However, it also points out candidate concerns regarding opaque decision-making and lack of feedback. The paper emphasizes the need for explainable AI (XAI) and human oversight to ensure ethical deployment and candidate acceptance, directly supporting SmartRecruit's human-in-the-loop and transparent evaluation approach.

- **Unveiling the Future of Artificial Intelligence in Talent Acquisition: A Bibliometric Analysis (2025)**

  Author: Nishanthi, B. et al.

  Publication: ScienceDirect.

  Summary: This bibliometric study analyzes extensive scholarly literature on AI-driven talent acquisition to identify emerging research trends, influential technologies, and existing gaps. The findings reveal a transition from basic automation to advanced AI techniques such as NLP-based resume parsing, predictive analytics, and AI-powered assessments. The study also highlights challenges related to ethical governance and real-world adoption, guiding SmartRecruit's roadmap toward scalable, ethical, and future-ready recruitment solutions.

- **Reimagining Recruitment: Traditional Methods Meet AI Transformation (2023)**
  Author: Rukadikar, A. et al.
  Publication: Cogent Business & Management.
  Summary: Rukadikar et al. explore the integration of AI technologies with traditional recruitment practices, demonstrating how AI enhances efficiency, reduces time-to-hire, and improves candidate quality. The study emphasizes that while AI is effective in screening and assessment, it has limitations in evaluating soft skills and organizational fit. This reinforces the need for a hybrid recruitment approach, aligning with SmartRecruit's design philosophy of combining AI insights with recruiter-driven decision-making.

- **The Role of Artificial Intelligence in Recruitment Process: Opportunities and Challenges (2021)**
  Author: Al-Alawi, A. I. et al.
  Publication: IEEE Access.
  Summary: This paper provides a balanced analysis of AI adoption in recruitment, outlining benefits such as automation, scalability, and data-driven decision-making alongside challenges including algorithmic bias, lack of interpretability, and legal compliance risks. The authors propose mitigation strategies such as bias-aware model development, transparency mechanisms, and periodic audits. These recommendations directly inform SmartRecruit's ethical AI framework and compliance-oriented system design.

- **A Review of Artificial Intelligence Based Platform in Human Resource Recruitment Process (2021)**
  Author: Saad, M. F. M. et al.
  Publication: IEEE Access.

Summary: This review examines existing AI-powered recruitment platforms and their functionalities, including resume parsing, candidate ranking, and automated interviews. The study highlights usability, integration with HR systems, and measurable outcomes as critical factors for adoption. It emphasizes that AI tools should support and enhance HR workflows rather than replace them, influencing SmartRecruit’s modular architecture and recruiter-centric dashboard design.

- **Impact of Artificial Intelligence on Recruitment and Selection Process (2021)**

  Author: Hemalatha, A. et al.

  Publication: IEEE Xplore.

  Summary: Hemalatha et al. evaluate the impact of AI technologies such as NLP, automation, and analytics on recruitment efficiency and scalability. The study reports significant reductions in screening time and improved candidate-job fit accuracy. It also highlights the increasing role of online assessments and automated evaluation mechanisms, justifying SmartRecruit’s implementation of ATS-based resume scoring, skill matching, and automated assessments.

- **Applicants’ Perception of Artificial Intelligence in the Recruitment Process (2023)**

  Author: Horodyski, P. et al.

  Publication: ScienceDirect.

  Summary: This research focuses on applicant attitudes toward AI-enabled recruitment systems, finding that candidates appreciate faster processing and objective evaluations but expect transparency, feedback, and human interaction. The study warns that poor communication may negatively affect employer branding. These insights guide SmartRecruit’s candidate-facing features, ensuring clear workflows, transparent evaluation stages, and enhanced communication.

- **Unveiling the Future of Artificial Intelligence in Talent Acquisition: Ethical Perspectives (2024–2025)**

  Author: Multiple Authors.

  Publication: Scopus Indexed Journals.

  Summary: This article addresses ethical challenges in AI-driven recruitment, including bias mitigation, accountability, fairness, and transparency. It proposes governance frameworks for responsible AI adoption and compliance with labor regulations. The study forms a foundational basis for SmartRecruit's ethical AI strategy, emphasizing explainability, auditability, and responsible handling of candidate data.

- **AI and Automation in Recruitment: Enhancing Candidate Matching and Selection Efficiency (2024)**
  Author: Multiple Authors.
  Publication: Scopus Indexed Journals.
  Summary: This study explores the application of machine learning and NLP techniques to improve resume-job matching accuracy by analyzing semantic similarity between candidate profiles and job descriptions. The findings demonstrate reduced hiring time, improved selection accuracy, and increased recruiter productivity, supporting SmartRecruit's objective of automated candidate screening and ranking.

- **Intelligent Recruitment Platforms: Design and Implementation Challenges (2024)**
  Author: Multiple Authors.
  Publication: IEEE / Scopus Indexed Journals.
  Summary: This paper discusses key architectural, technical, and ethical challenges in developing intelligent recruitment platforms, including data privacy, system scalability, AI explainability, and HR system integration. The authors emphasize secure data storage, modular system design, and continuous AI monitoring, which directly influence SmartRecruit's platform architecture, security policies, and long-term maintainability.

## 2.3 Existing Systems

- **Traditional Applicant Tracking Systems (ATS)**

Examples: Taleo, SAP SuccessFactors, Workday ATS

The traditional Applicant Tracking Systems (ATS) are common systems that are used by organizations to maintain job applications, candidate records, and recruitment processes in a centralized and organized manner. These systems largely work based on rule-driven filtering system and keyword-matching algorithm to narrow down the list of candidates based on pre-set job requirements like educational level, technical expertise, level of experience and certification. Applicants have their resumes filtered by recruiters for whom certain screening criteria is set and those resumes that do not fit in the pre-set criteria are automatically filtered at the first stage of selection. The method aids organizations to effectively manage high numbers of applications and minimizes manual work during the initial screening phases. Also, ATS platforms feature workflow automation like application tracking, interview scheduling, candidate status management and communication templates that will help to better organize and document the recruitment process.

Nevertheless, even though they have operational benefits, there exist some limitations in traditional ATS platforms which influence the efficiency and equity of evaluation of candidates. Because these systems are highly dependent on a literal match of the keywords and not the contextual interpretation, they are usually ineffective at recognizing semantically relevant skills in different terms or formats. Consequently, there is a likelihood that potentially qualified applicants can end up being rejected due to the initial screening process. Moreover, the majority of the traditional ATS systems lack embedded skill testing tools like coding tests, MCQ-based tests, or AI-enhanced candidate ranking systems. Neither do they have intelligent analytics dashboard that give a more in-depth look at the suitability of candidates and hiring trends. Therefore, recruiters have to rely on other external assessment and performance evaluation tools, which means that the workflow is disjointed, and choices are more complicated. These constraints indicate the necessity of a higher-level AI-based recruitment solution that facilitates semantic analysis, combined evaluations, and data-driven recruiting insights, to enhance the accuracy and efficiency of the recruitment process.

- **Job Portals:Examples: LinkedIn, Naukri, Indeed**

Job portals are mainly used as online sources of job advertisement and sourcing of applicants by linking employers to a big pool of potential employees in various industries and geographical areas. Such platforms allow recruiters to post job advertisements, specify eligibility requirements and applicants to submit their applications using standardized workflow techniques. Centralized dashboards allow candidates to create profiles, upload resumes, job application tracking, and apply to various opportunities. Also, job portals offer the simplest level of filtering and search options, according to parameters like location, experience level, educational qualification, job role, and keyword relevancy, which assists recruiters to conduct initial-level shortlisting of candidates. Their extensive user base and accessibility drastically enhance organizational visibility and reach, which makes them a viable instrument in attracting a variety of talent. Nevertheless, even though job portals are advantageous in efficiency in their candidate sourcing, they usually do not have sophisticated analytical features needed to support the intelligent decision making in recruitment. The vast majority of platforms do not conduct any deep semantic resume analysis and assess the appropriateness of candidates beyond basic algorithmic matching of keywords and profile filtering. Consequently, recruiters have to spend hours or days manually sifting through vast amounts of applications or extracting candidate data to third-party Applicant Tracking Systems (ATS) and evaluation tools to evaluate them further. Also, these portals usually lack built-in skills-based evaluations, artificial intelligence-based candidate ranking systems, and real-time performance analytics to enable structured hiring processes. As a result, whereas job portals are significant in enhancing the reach of a candidate and the ease in accessing recruitment, they do not provide much assistance in automated screening, competency assessment, and overall recruiting automation. These restrictions underscore the importance of smart recruitment tools that can combine sourcing features with AI-powered evaluation and assessment processes to improve hiring effectiveness and decision accuracy.

- **Online Assessment Platforms Examples: HackerRank, Mercer Mettl,**

**HireVue :** Online assessment tools are designed to test competency of a candidate using systematic testing programs like Multiple Choice Question (MCQ) tests, code cracking, technical simulations, and AI-enhanced video interviews. With these platforms, recruiters can gauge the theoretical understanding, programmer skills, logical thinking capacity, and problem solving of applicants in a regulated online setting. Webcam surveillance, browser activity monitoring, screen-recording, behavioral analysis are also many of the AI-assisted proctoring features added to many modern assessment systems to ensure examination integrity and decrease the chances of malpractice during remote assessments. They also enable automated scoring, performance analytics, and role-specific test customization, which enable recruiters to quickly find technically qualified candidates based on measurable performance indicators.Although they can be effective in evaluating candidates, the online assessment systems are generally one-off solutions that are not well coupled with the resume screening or Applicant Tracking System (ATS) processes. They do not typically have smart resume parsing, semantic skill extraction, and automated resume-job matching functions needed to perform initial candidate screening. Consequently, recruiters may need to manually bulk transfer shortlisted applicants out of ATS systems or job portals to assessment systems, resulting in disjointed workflows and adding additional administrative burdens. Moreover, the majority of evaluation systems offer little to no support in centralized candidate comparison dashboard, hired analytics, or life cycle recruitment. This division of the sourcing, screening and assessment processes makes the recruitment more inefficient, and it is hard to ensure consistency in the evaluation stages. Thus, by incorporating assessment functionalities into a smart recruitment support system, workflows can be greatly simplified, evaluation continuity enhanced, and more accurate and data-driven hiring decisions made possible.

**Table 2.3** Comparative study of Existing systems

| Feature | Traditional ATS | Job Portals | Assessment Platforms | Smart Recruit |
|---|---|---|---|---|
| Resume Evaluation | Keyword-based | Basic filtering | ✕ | NLP-based similarity scoring |
| Skill Assessment | ✕ | ✕ | ✔ | ✔ |
| Proctoring | ✕ | ✕ | Limited | AI-based proctoring |
| Unified Careers Page | ✕ | ✕ | ✕ | ✔ |
| Decision Insights | Limited | Limited | Test-only | Comprehensive analytics |

In the comparative analysis, the current recruitment systems offer partial solutions, either concentrating on resume storage and filtering by keywords (Traditional ATS), sourcing applicants (Job Portals) or skill measurement and proctoring (Assessment Platforms). Such systems are usually isolated and do not have a cohesive structure that would offer an overview of how well a candidate fits in a systematic and systematic way. Consequently, the recruiters usually need to use various platforms to fulfill the different steps of the recruitment process and this complicates operations, decreases workflow efficiency and can create discrepancies in the results of the evaluation processes. Moreover, it is hard to conduct a comparative evaluation of different candidates on multiple parameters including technical skills, experience relevancy, assessment performance, and role alignment due to the lack of integrated analytical support. This piecemeal system denies organizations the capability of making timely and informed hiring decisions.

The Careers Page: Smart Recruit will overcome these shortcomings by incorporating the inherent advantages of the traditional recruitment tools into one intelligent platform that will allow end-to-end management of the recruitment process. It incorporates AI-based resume analysis, similarity scoring on ATS, skills and experience matching using NLP and structured online tests with real-time proctoring system to enhance fairness and authenticity in the process of evaluation. Moreover, the platform has also added recruiter decision-support dashboards that visually reveal insights of candidate performance, ranking trends, and suitability metrics to facilitate quicker and more consistent shortlisting decisions. The proposed system facilitates clear, objective, and data-driven recruitment strategies combined with providing the much-needed human control in the ultimate hiring processes, using Natural Language Processing to perform semantic resume-job matching and using computer vision to monitor assessment integrity.

The proposed system is supposed to be an intelligent, scalable and reliable recruitment system which will greatly benefit the accuracy of candidate evaluation, increase the credibility of assessment and assist recruiters with the help of actionable analytical information. The system simplifies the work of individuals by integrating resume assessment, automated grading systems, similarity scoring systems, and performance analysis tools into a single careers page interface, thus decreasing the amount of manual work and reducing the reliance on various outside tools. The platform also enhances the transparency of recruitment processes by ensuring the evaluation steps are well-organized and communication between applicants and recruiters is more convenient during the hiring process. In general, the proposed Smart Recruit system is a step in the right direction of creating a more effective, stable, and reliable recruitment ecosystem that will benefit organizations and job seekers by enhancing the quality of decisions, minimizing the time spent on the processing, and boosting confidence in AI-based recruitment settings.

## 2.4 Technology Survey

The application combines various recent technologies in the front-end, back-end, artificial intelligence modules, assessment infrastructure and database layers to create a scalable and smart recruitment support platform. The system architecture is modular whereby each element performs a particular task like resume processing, assessment execution, similarity scoring, candidate analytics and recruiter decision support. These modules interact via secure REST based interfaces, allowing interaction between system elements with ease, performance, scalability, and security throughout the recruitment lifecycle. : -

- **Backend Development:** The system backend is created with the help of Node.js and Express.js, which is the central application server and handles the system logic and coordinates communication between the frontend interface and the AI-based resume analysis modules and the database layer. It manages basic services like user authentication and authorization, job posting, resume uploading, ATS similarity scoring requests, assessment scheduling and execution workflow, proctoring activity logging, and recruiter dashboard data aggregation. The backend provides secure RESTful

APIs that guarantee the systematic exchange of data among components of the system and ensure reliability and scalability. Mechanisms of role-based access control are also introduced to ensure that candidates and recruiters have distinct permissions that reduce the risk of sensitive recruitment data being accessed by unauthorized individuals and also prevent unauthorized access to restricted system functionality

- **Frontend Interface:** React.js is used to create an interactive and convenient web interface with HTML, CSS, and JavaScript that offers interactive and user-friendly careers page to both candidates and recruiters. The frontend allows the candidates to search through the job openings, submit resumes, follow-up, and take online tests using a user-friendly workflow. Recruiters receive special dashboards with rankings of candidates, similarity, assessment, and visual analytics that aid in decision-making. The interface is developed based on the principles of usability and accessibility to provide easy navigation, responsiveness to layouts on all devices, and an intuitive interface with the recruitment processes. Also, real-time updates of application progress and assessment results are provided because of asynchronous communication with backend APIs.

**Resume Parsing and NLP Analysis :** The uploaded resume files in PDF or DOC formats are analyzed with Python-implemented Natural Language Processing (NLP) modules to extract structured data out of unstructured resume text. The preprocessing pipeline involves tokenization, stop-word elimination, normalization and key word extraction to enhance the quality of the data prior to similarity calculation. The system uses the Jaccard Similarity and Cosine Similarity methods to compare candidate resumes with the job descriptions to provide the appropriate ATS-based similarity scoring and skill relevance assessment. Such similarity measures allow semantic matching over straight keyword filtering, thus enhancing accuracy of candidate-job match, and minimizing the chances of eliminating qualified candidates as a result of terminology differences.

**Online Assessment and Proctoring:** The assessment module facilitates

MCQ based assessment and coding assessment that aims to assess not only theoretical knowledge but also practical proficiency of solving problems. Real-time proctoring mechanisms are used to implement fairness and authenticity in remote assessment based on computer vision practices. These consist of face detection using a webcam, detecting more than one face in the frame and monitoring of browser tabs to detect suspicious activity during test session. The proctoring logs are stored safely and can be reviewed by the recruiters to achieve transparency and accountability in the evaluation of the assessment but retain human oversight in the final decision-making.

- **Database :** The main database system used is MongoDB to store and process structured and semi-structured data related to recruitment. It has databases of user profiles, resumes, job ads, application status, similarity rating, assessment results, and proctoring activity logs. MongoDB uses a document-oriented architecture, which provides a very flexible schema design to enable the database to store various types of recruitment data without strict relational restrictions. This design enhances scalability of the system, faster query response to dashboard analytics, and effective storage of changing candidate data during the recruitment lifecycle. Besides, safe access control systems provide the integrity of data, privacy, and long-term reliable storage of the information connected with recruitment.
- **System Integration And Scalability Consideration** : Combining Node.js backend services and React-based frontend elements with Python-based NLP modules, computer vision-based proctoring solutions, and MongoDB storage architecture forms a unified and scalable recruitment system that could handle the end-to-end recruiting processes. The modular architecture enables single components, including similarity scoring engines, assessment modules, or analytics dashboards to upgrade without impacting the overall system stability. This makes the platform flexible to any other future improvements like ranking of candidates by machine learning, explainable AI scoring, and advanced recruitment analytics to plan the organizational workforce.

# Chapter 3

# HARDWARE AND SOFTWARE REQUIREMENTS

## Development Environment

## 3.1 Hardware Requirements

*Table 3.1: Hardware requirements*

| Hardware Specification | |
|---|---|
| **Specification** | **Desired Value** |
| Processor | AMD Ryzen 5 4600H |
| RAM | 8GB Minimum |
| Hard Disk | 500 GB Minimum |

## 3.2 Software Requirements

*Table 3.2: Software requirements*

| Software Specification | |
|---|---|
| **Specification** | **Desired value** |
| Operating System | Windows 10 (64-bit) / Windows 11 |
| Frontend Tool | React 18.x |
| Backend Tool | Node.js 18 LTS, TensorFlow.js 4.x |
| Database | MongoDB 6.x |
| Development Tool | VS Code 1.85+ |

# Chapter 4

# SOFTWARE REQUIREMENTS SPECIFICATION

## 4.1 System User

- **Candidate**

  A candidate refers to a person, who wants to get a job opportunity by means of the Careers Page platform, and is one of the main stakeholders of the Smart Recruit system. The candidates engage with the system by creating and maintaining their personal profiles, posting resumes in their supported formats, searching through the available positions and making applications to relevant positions. The site allows the candidates to take standardized online tests, such as MCQ-based exams and coding tests, which are meant to assess both the theoretical knowledge and the ability to solve practical problems applicable in the work context of the job being applied to. By using ATS-based resume similarity scoring which is based on Natural Language Processing (NLP) algorithms, job seekers are given an objective idea of how well they fit on job descriptions, enhancing their perception of job requirements. In addition to evaluation support, the system enhances candidate experience by providing structured workflows, transparent assessment procedures, and real-time application status tracking throughout the recruitment lifecycle. Applicants can also track the process at various recruitment phases including screening of resumes, completion of assessment and selection of shortlisted candidates. The platform can also offer performance based insights and feedback indicators to enable the candidates know their strengths and areas they should improve on. The system enhances trust, engagement and confidence among the applicants by being clear on evaluation criteria and being fair through monitored assessment environments, which foster active involvement in the recruitment process.

- **Recruiter**

  A recruiter is a management user of the Smart Recruit platform who manages and oversees recruitment. The recruiters engage with the system to design and post job advertisements, outline role-specifications, screen the applications of candidates, and track the recruiting process using centralized dashboards. The system assists recruiters with automatic analysis of posted resumes based on ATS-based similarity scoring and NLP-based skill matching methods, allowing them to easily find those c candidates whose qualifications match those of a job. Also, by using online testing, recruiters are able to assess the performance of the candidates and they can also access proctoring logs produced via real-time monitoring systems, in order to identify authenticity and fairness when evaluating the candidate. The platform also supports recruiters through automated candidate ranking systems, visual analytics, which show insights into candidate suitability, assessment performance trends, and status of the hiring pipeline. These analytical dashboards save on manual screening and allow recruiters to carry out comparative analysis of various applicants with structured performance metrics.Although intelligent automation is used, the system maintains recruiter control over the ultimate hiring, so that human judgment remains at the center of hiring candidates. Altogether, the recruiter module is more efficient in its operations, better in evaluation consistency, and data-driven decision-making, without losing flexibility, or transparency all along the recruitment workflow.

## 4.2 Functional requirement

These are the functional requirements.

- **Candidate Module**

**FR1: User Registration, Login, and Profile Management**
The system will enable candidates to sign up with authentic credentials like email and password under a secure authentication system. The system will facilitate validating logins and management of sessions that will provide authorized access to candidate specific features. The job applicants will be in a position to create, update, and maintain their accounts with personal information, education, technical expertise, credentials, and work experience. The profile attributes will be stored safely and utilized in resume analysis, similarity ranking and clever job-position matching. The system will also enable candidates to periodically update profile information so as to ensure accuracy and better performance matching.

**FR2: Upload Resume for ATS Analysis**
The system will enable candidates to post resumes in formats supported by it like PDF files or DOC files via Ceres Page interface. Resumes uploaded will undergo parsing in the form of ATS based parsing mechanisms to obtain structured data such as skills, years of experience, qualifications and keywords in terms of

job description. The information obtained will be saved safely and will be employed in similarity scoring and recruiter-side assessment. The system will confirm compatibility of file formats and successful upload confirmation to candidates.

**FR3: View ATS Similarity Score and Improvement Suggestions**
The system will produce an ATS similarity score through a comparison of candidate resumes with job descriptions based on techniques of NLP-based similarity computation, including a cosine similarity and a keyword overlap analysis. The candidates will be capable of seeing their similarity rating via the dashboard interface as well as automatic recommendations to optimize resume fit with the position applied to. Such recommendations can involve absent skills, suggested keywords, or formatting suggestions that can aid in candidate-job fit.

**FR4: Attempt MCQ / Coding Assessments with Live Proctoring**
The system will provide online testing based on the roles where MCQ tests and coding tasks will be offered to test the theoretical knowledge and practical skills. In the testing session, real time proctoring systems will be enabled to track the activity of the candidates through webcam face detection and browser monitoring systems. Such monitoring systems will guarantee validity of involvement and create equity in measurement sessions.

**FR5: Receive Real-Time Alerts for Proctoring Violations**
The system will inform the candidates as soon as the violation of proctoring is detected during testing. The violations could be the lack of face detection, the possibility to see more than one face in the camera frame, or even between the tabs in the browsers. Live notification will be used to assist candidates to rectify behavior and proceed with tests under compliant circumstances thus facilitating transparent and fair testing procedures.

**FR6: Search and Apply for Available Job Postings**
The system will enable candidates to search job postings via the Careers Page interface and apply filters to filter job postings by job role, skills required, and experience level, and application deadlines. Applicants will be in a position to see job descriptions and apply on the platform. The system will verify the application submission and show application history in the candidate dashboard.

**FR7: Track Application Status and Assessment Results**
The system will allow the candidates to monitor the advancement of their job applications by various recruitment processes including resume screening, assessment completion, shortlisting and the final decision outcomes. Applicants will additionally have the ability to see scores of assessments and performance indicators on their own dashboard, enhancing transparency and interaction in the recruitment lifecycle

- **Recruiter Module**

**FR8: Create and Manage Job Postings**
The system will enable recruiters to create, update and delete job postings using a special recruiter interface. The recruiters will be in a position to specify role specifications including the skills required, level of experience, level of education and job duties. The ATS similarity scoring engine will use these job descriptions to determine the compatibility of candidates with posted positions.

**FR9: View Candidate List with ATS Ranking and Filters**
The system will provide a prioritized list of candidates to fill a job post in terms of ATS similarity scores and assessment performance metrics. Recruiters will be in a position to filter lists of candidates with set parameters like the level of skills match, assessment scores, level of experience, as well as the relevance to the qualification. This ranking system will facilitate effective comparisons of candidates and well organized shortlisting.

**FR10: Review Quiz Results and Proctoring Reports**
The system will give the recruiters access to candidate evaluation performance data such as MCQ scores, coding test scores, and completion percentages. Furthermore, recruiters will be capable of going through proctoring activity reports pointing out violations detected, including multiple face detection, camera inactivity, or suspicious browser activity. These reports will help recruiters in determining the authenticity of assessments prior to the final evaluation.

**FR11: Shortlist or Reject Candidates with Automated Email Notifications**
The system will enable the recruiters to shortlist or reject the candidates based on the dashboard interface after examining the similarity scores of the resume, assessment performance, and proctoring reports. Email notification: Automated email messages will be created and sent to candidates about recruitment decisions in a reasonable and transparent way. These alerts will enhance the effectiveness of communication between the recruiters and the applicants.

**FR12: Visualize Recruitment Metrics through Analytics Dashboard**
The system will offer recruiters an analytics dashboard which shall show recruitment statistics including number of applicants per job role, similarity score distribution, statistics on shortlisted candidates, assessment performance summary, hiring pipeline status. These visual insights will be used to aid in data-driven recruitment decision-making and assist recruiters in tracking overall hiring efficiency when hiring across various job postings.

## 4.3 Non-Functional requirement

The quality attributes, performance expectations, security standards and the operational characteristics of the Smart Recruit Careers Page platform are determined via non-functional requirements. These conditions will make sure that the system is efficient, secure, reliable and consistent in its operation as it supports intelligent recruitment processes among the various user roles.

- **Performance:**

  The system aims at offering effective processing and real-time responsiveness to all key recruitment workflows. The ATS similarity scoring engine scores resume-to-job compatibility scores in about 5 seconds after resume upload, allowing candidates and recruiters to gain instant analysis. The assessment module aids continuous implementation of MCQ and code tests with minimum latency. Also, real time proctoring subsystem has a face detection and monitoring capability that is web-based to maintain constant monitoring during assessment and has a frame rate of around 15-30 frames per second, which does not interfere with system utilization. Asynchronous request handling implementations in the form of backend services also enhance the response time in dashboard analytics, candidate ranking retrieval, and job application tracking activities.
- **Scalability:**

  The platform is built on a scalable architecture that has the potential to handle a high number of simultaneous users, such as candidates undergoing assessments and recruiters handling a variety of job listings at the same time. Asynchronous processing using Node.js and document-oriented storage model offered by MongoDB allow effectively processing large volumes of resume uploads, similarity scoring requests, and assessment activity logs. The system architecture is compatible with cloud environments, with the ability to horizontally scale services like resume parsing modules, assessment engines, and analytics dashboards. The platform is scalable to organizations of all sizes such as startups, educational and enterprise level recruitment environments.

- **Security:**

  Security is an important factor towards securing sensitive information of the candidates and recruiters during the recruitment process. To ensure secure user login sessions and controlled access to the system, the system has JWT-based authentication mechanisms. Role-based authorization policies help to avoid unauthorized operations within the platform by restricting access to recruiter-only or candidate-only features. Encrypted storage mechanisms are used to keep sensitive data like resumes, assessment results and proctoring activity logs confidential and intact. Also, communication protocols that are secure REST APIs are used to ensure that the information shared between the frontend interfaces, back-end services, and AI processing modules is not intercepted or modified by the attackers.

- **Usability:**
  The Careers Page interface is created to have an easy, user-friendly and interacting interface to the applicants and recruiters. The frontend interface that is built on react.js provides a seamless navigation through job listing, resume upload modules, assessment environment and recruiter dashboards. Good workflow organization enables applicants to follow-up application and assessment status quickly and recruiters can easily maintain job advertisements and applicant reviews on centrally located dashboards. The responsive design facilitates usability on various devices such as desktops, tablets and mobile browsers, and hence accessibility among various levels of users with varying technical skills.
  **Reliability:**
  The system will be able to offer 24/7 availability and high operational reliability of the system to ensure that there is continuous service in recruiting services. Secure session management and error-handling mechanisms will make sure that even under different network conditions, assessment sessions are stable. The backend architecture maintains uniform response to resume processing requests, assessment execution workflows, and dashboard analytics retrieval without breaking down the service. Periodic database backup plans and supervisory mechanisms are also used to increase the reliability of the system, eliminating the chances of data loss and continuity of recruitment processes.
- **Accuracy:**
  The site provides very high evaluation accuracy by combining Natural Language Processing (NLP)-based similarity scoring models that utilize semantic relationship between resumes and job descriptions as opposed to just using keyword matching. This enhances fit between candidates and jobs and minimizes

the wrong screening of qualified candidates. The proctoring subsystem also employs computer vision to deliver effective face detection, multiple face identification and monitoring of browser activity, in assessments. Such mechanisms give greater credibility to the assessment and make sure that all candidates undergoing remote recruitment processes are fairly evaluated.

- **Maintainability:**
  The Smart Recruit platform follows a modular and service-oriented architecture, enabling independent development, testing, and enhancement of system components such as resume parsing modules, assessment engines, analytics dashboards, and authentication services. It is a modular structure, which eases the process of debugging, enhances the adaptability of systems, and helps to add new features in the future, e.g. explainable AI scoring systems, predictive hiring analytics, or sophisticated recruiter recommendation engines. Adequate documentation and standardized API communication also enable future maintenance and scalability of the system without interfering with the current system features.

# Chapter 5

# SYSTEM DESIGN

## 5.1 Architecture Diagram

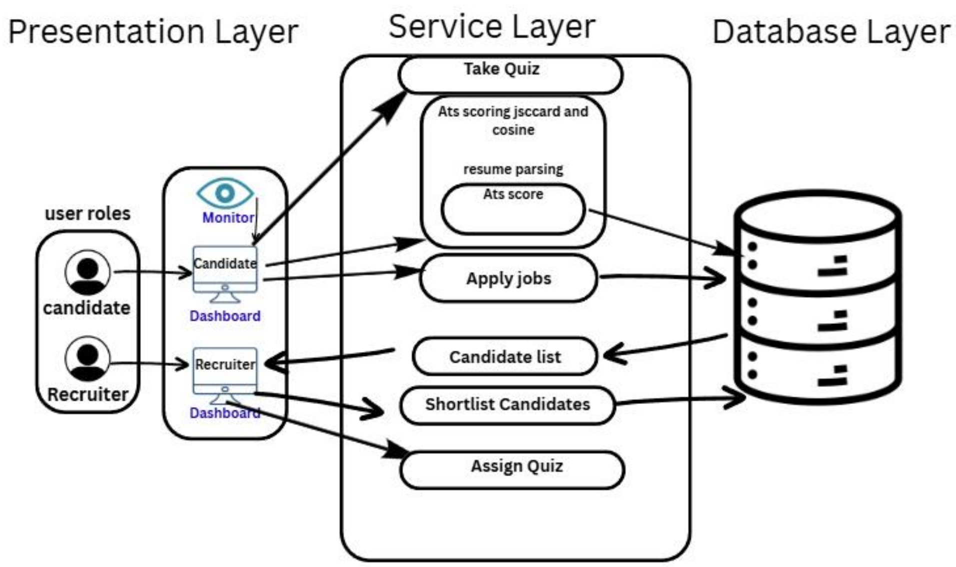



*Figure 1 Architecture Diagram of Careers Page: A Smart Recruit*

**Architecture Layers Description**

**1. Presentation Layer (Frontend UI)**

- Offers web interfaces between Candidates and Recruiters.Enables resume upload, job
- Facilitates resume upload, job application, assessment, dashboards and analytics.
- Built using modern frontend technologies for responsive and user-friendly interaction.

**2. Application Layer (Backend APIs)**

- Plays the role of the master controller of the system.Acts as the central controller of the system.
- Deals with authentication, job processing, resume, tests, and notifications.
- interacts with the database and AI services using secure APIs.

**3. AI Layer (ATS & Proctoring Engine)**

- Scores ATS resume based on NLP methods (Jaccard and Cosine Similarity).

- Elicits competencies and appraises experience.
- Performs real-time proctoring, with face recognition, tab switch, and multiple face warnings during testing.

**4. Data Layer (Database & Storage)**

- Stores user profiles, resumes, job postings, ATS scores, assessment results, and proctoring logs.
- Ensures secure and reliable data persistence.

## 5.2 Context Flow Diagram

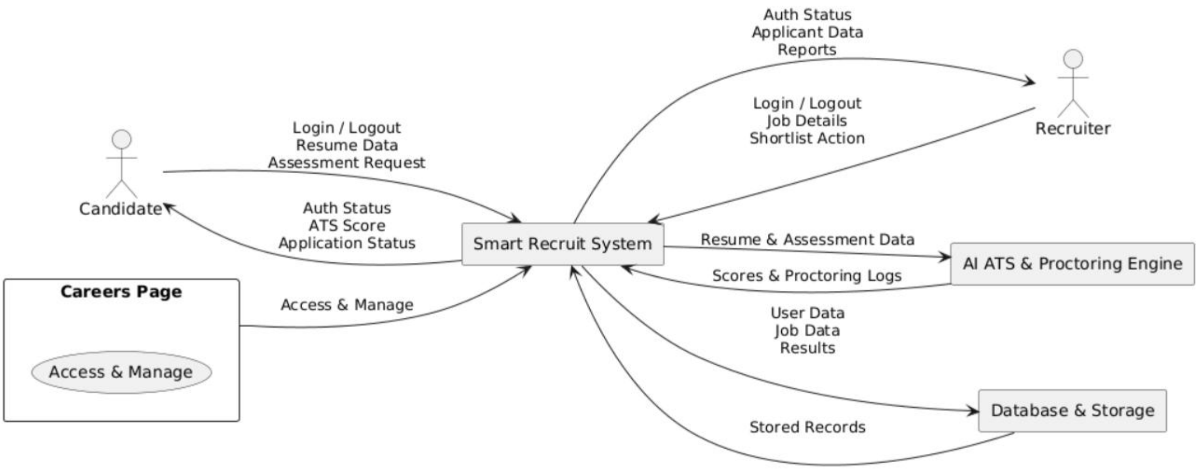



*Figure 2 Context Flow Diagram of Careers Page: A Smart Recruit*

**Context Description**

The Smart Recruit System is an intelligent recruitment support system, which enables communication between two main external parties; Candidates and Recruiters. The system is a centralized interface that provides management of resume processing, assessment operations, similarity scoring, and recruitment analytics through integrating processing services in the back-end, AI-based evaluation modules, and secure data-storing mechanisms. The platform is designed to make sure that the communication and coordination between users and system components throughout the entire recruitment lifecycle is efficient through organized workflows and automated decision-support features.

- **Candidates** interact with the system to:
  - The Smart Recruit platform allows candidates to engage with the careers page interface to carry out different activities pertaining to recruitment in

an organized and clear way.

- The system enables candidates to: Granted access through authenticated credentials to personalized dashboards and recruitment features.
- Establish and maintain professional accounts with personal details, education, skills, and certification and work experience, which are used to match with jobs.
- Post resumes in compatible formats like PDF or DOC to use the automated processing of resumes based on similarity scoring systems using automated tools like ATS. See ATS similarity scores generated by comparing resumes and job descriptions with NLP-based methods, and allowing the candidates to appreciate how they match with a particular position.
- Undergo proctored online examinations, such as MCQ-based tests and coding tasks that aim to test theoretical knowledge and practical ability. Get on-the-fly notifications when assessments are conducted in case of any violation of proctoring like face detection or tab-switching.
- Monitor the status of applications at various levels of the recruitment process such as screening of resumes, completion of assessment, making shortlisting decisions and final selection results using the candidate dashboard.
- These capabilities enhance higher engagement with the candidates by offering transparency, systematic work flow, and prompt feedback on the evaluation process.

- **Recruiters**

  Recruiters use the Smart Recruit site in the form of a specialized recruiter dashboard that can facilitate effective organization of the recruitment processes and candidate review processes.

  - The system allows recruiters to: Post, edit and control job advertisements with job-specific criteria like skills and experience level and qualification. Browse lists of ATS-ranked candidates created automatically by their similarity scores and job-role fits.
  - Sift and sort the applicants based on predefined evaluation criteria such as performance on tests and relevancy of experience. Check assessment

outcomes and proctoring report to identify the authenticity of the candidate and provide fairness during the evaluation.

- Filter or dismiss candidates right on the dashboard using insights on the data. Track recruitment analytics dashboards, which give visual summaries of the applicant distribution, the selection process, and the hiring pipeline performance.
- Such features help to decrease the amount of manual screening and assist recruiters in making hiring decisions which are quicker, more consistent, and evidence-based.

**The Smart Recruit System:**Smart Recruit system is the central processing unit which facilitates communication between users and AI modules as well as the database infrastructure. Its responsibilities include: ATS similarity scoring techniques, which utilize Natural Language Processing methods to process resumes and job descriptions, were used to support processing. Calling on AI services to extract semantic skills, match candidates and roles, and proctoring of real-time assessment by means of computer vision monitoring. Handling assessment procedures, such as scheduling, execution, monitoring and storing of results. Logging proctoring activity and producing violation notification, which the recruiter can review. Accessing and storing recruitment information safely out of the database such as candidate profiles, job advertisement, assessment results, similarity indexes, and updates about application status. Delivering analytical information in the form of dashboards that facilitate recruiter decision-making and enhance visibility of recruitment pipeline.

# Chapter 6

# DETAILED DESIGN

## 6.1 Use Case Diagram

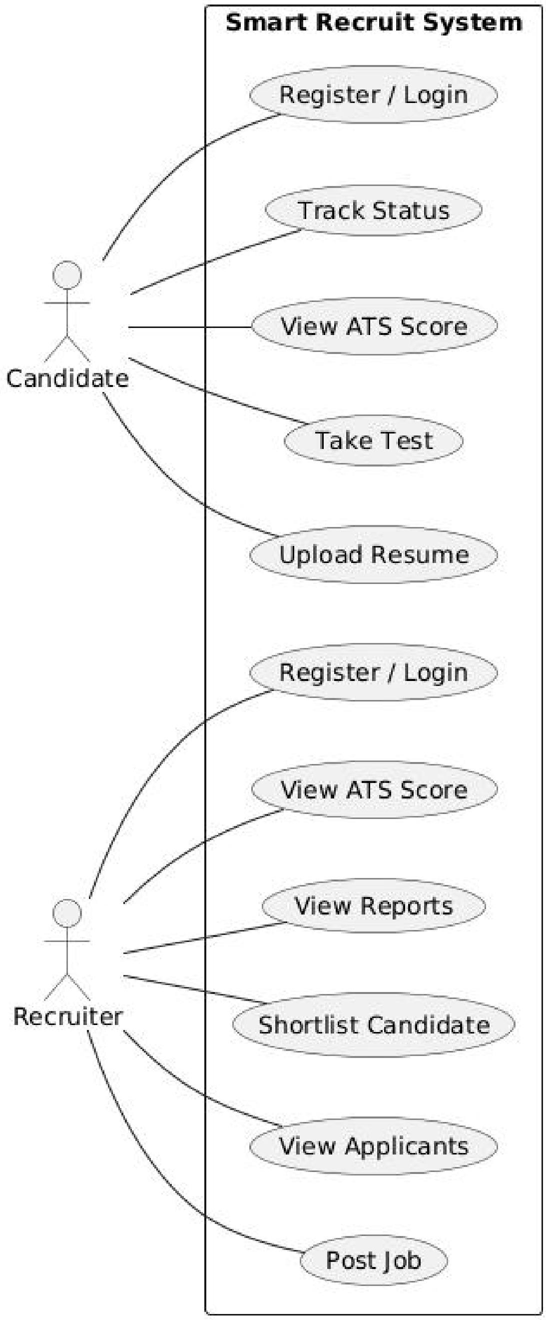



*Figure 3 Use case Diagram of Careers Page: A Smart Recruit*

The Use Case Diagram shows how the system users interact with the Smart Recruit platform focusing on how the various actors use the system functionalities to carry out recruitment related operations. The diagram represents the functional scope of the system by identifying primary users and their respective responsibilities within the recruitment workflow. The Smart Recruit platform serves two key participants: Candidate and Recruiter, each having an interface with the system that enables the participation in the activity according to the role-specific features of the system that are involved in the submission of the resume, assessment, similarity score, candidate evaluation, and recruitment decision-making. The use case diagram assists in visualizing how the system supports communication between the system modules and the user as well as the well-structured workflows throughout the recruitment lifecycle. It also shows the integration of intelligent automation features like ATS similarity scoring, online assessments and recruiter analytics dashboards into the interaction model to enhance the efficiency of the recruitment process and transparency

**Actors**

- **Candidate :** A candidate is a person who wants to be offered an opportunity to be employed using the Smart Recruit careers page. Applicants engage with the platform to design profiles, post resumes, complete tests, and see similarity scores, as well as track the application status at various recruitment phases.
- **Recruiter :** An authorized organizational user who takes care of job posting, screening of applications, assessment scoring, and analysis of similarity rankings of ATS and shortlisting or rejection is called a recruiter using the recruiter dashboard interface.

**Use Cases**

**Candidate Use Cases**

- Register / Login The system enables applicants to safely enroll with authentic credentials and log in to get dedicated dashboards. Authentication guarantees access of features that are specific to the candidate like resume uploading, participating in an assessment and monitoring of their applications.

- Upload Resume Candidates can upload resumes in supported formats such as PDF or DOC through the careers page interface. ATS-based similarity scoring mechanisms are used to process the uploaded resumes that harvest pertinent data such as skills, experience, and job-role matching qualifications.
- View ATS Score Once the resume is processed, the applicants will have access to their ATS similarity score, which was created by comparing their resumes to job descriptions, using NLP. The similarity score makes the candidates know the level of their profile that matches the job requirements of a given job.
- Take Online Assessment Applicants have the opportunity to take part in role based online tests which are multiple choice question tests and coding tasks. These exams test the knowledge and the practical skills of problem solving. Real time proctoring measures are used to provide equity in the execution of the assessment.View Application Status Using their dashboard, the candidates are able to monitor their application status through various recruitment phases including screening of their resumes, completion of assessments, shortlisting, and final selection.

**Recruiter Use Cases**

- Register / Login To access recruiter-specific features, including job posting management, candidate evaluation dashboards, and recruitment analytics tools, recruiters can securely register and log in to the Smart Recruit platform.
- Post Job Recruiters are able to formulate and post job advertisements by defining job requirements in terms of technical expertise, level of experience, qualifications and duties. The ATS engine uses these job descriptions to determine the compatibility of the candidates.
- View Applicants The system gives the recruiters a methodical list of candidates to the job posting. Through the recruiter dashboard interface recruiters can view the candidate profiles, resumes and assessment participation status.
- View ATS Ranking Automatically generated rankings of applicants according to ATS similarity scores and assessment performance measures are available to recruiters. These rankings assist in effective comparison of the candidates and

recruiters can select the most appropriate candidates to be evaluated further.

- Shortlist / Reject Candidate Based on the similarity scores, assessment performance and proctoring reports, recruiters will be able to shortlist or reject a candidate.

## 6.2 Activity Diagram

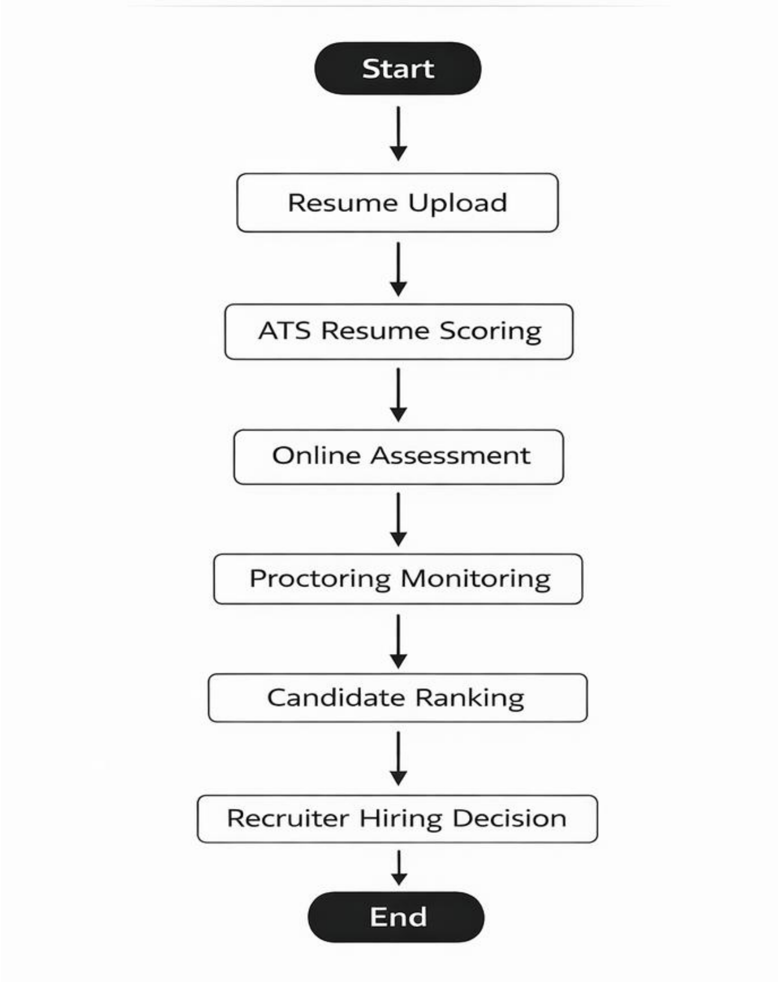



*Figure 4 Activity Diagram of Careers Page: A Smart Recruit*

The Activity Diagram shows the chronological flow of recruitment processes in the Smart Recruit system that begins with the interaction between a candidate with the careers page all the way to the ultimate hiring decision taken by the recruiter. It represents the step-by-step execution of system processes and highlights how resume evaluation, assessment monitoring, similarity scoring, and candidate ranking are integrated into a structured recruitment pipeline. The schematic assists in picturing the interaction between the components of the automated system and user interactions to promote transparent, efficient, and data-driven hiring decisions. The process of the activity is the interaction of the candidates with the Smart Recruit platform, AI-based evaluation modules, and recruiters. Every step in the working process helps to enhance the accuracy of the candidate-role matching, the integrity of the assessment, and assists recruiter decision-making with analytical insights.

**Activity Flow Description**

1. Candidate Uploads Resume When a candidate uses the careers page interface to upload his/her resume, the recruitment process starts. The system takes resumes in the supported formats like PDF or DOC and stores it in the database in a secure place. Resume data is then sent to the ATS processing module to be structured to extract candidate skills, qualification, certifications and experience information. The step is the basis of similarity scoring and other evaluation.
2. System ATS based resume scoring. Upon resume submission, the system uses Natural Language Processing (NLP)-based similarity algorithms, including Jaccard Similarity and Cosine Similarity to match the resume of the candidate to the job description. According to this comparison, the platform creates an ATS similarity score that represents the level of similarity between the profile of the candidate and the job role required. Candidates and recruiters can access the similarity score on their respective dashboards to enable open assessment.
3. Candidate Takes Online Assessment. Shortlisted candidates with the first similarity cutoff or with intentions to undergo further testing are invited to take role-based online tests. Such tests can comprise multiple-choice questions (MCQs) and coding problems that can test conceptual knowledge as well as technical problem-solving skill, which is applicable in the employment position.
4. Proctoring on-the-fly Monitors Assessment. The system provides real-time proctoring features during the assessment session to track candidate activity by using both face detection and browser tab-switch tracking technologies with the help of webcams. The proctoring engine recognizes breaches like lack of face recognition, several people in the camera view or changing browser tabs. The proctoring logs are captured and kept to be reviewed by the recruiters during assessment evaluation to promote fairness and authenticity in the evaluation.

5. System Ranks Candidates Upon the completion of assessments, the system combines evaluation parameters such as ATS similarity scores, assessment performance results, reports on proctoring activities to produce a ranked list of candidates. This ranking assists recruiters to quickly compare the performance of the applicants according to the structured performance measures and suitability indicators as per the job requirements.
6. Recruiter Decides to Hire. Lastly, recruiter peruse rankings of candidates, resume similarity scores, testing results, and proctoring reports using the recruiter dashboard interface. With these insights, recruiters may shortlist or dismiss candidates and continue with additional recruitment processes like interviews or final selection. Candidates are then informed about the result of their applications by automated notices and this makes the recruitment process transparent and communicates effectively.

## 6.3 DATABASE DESIGN

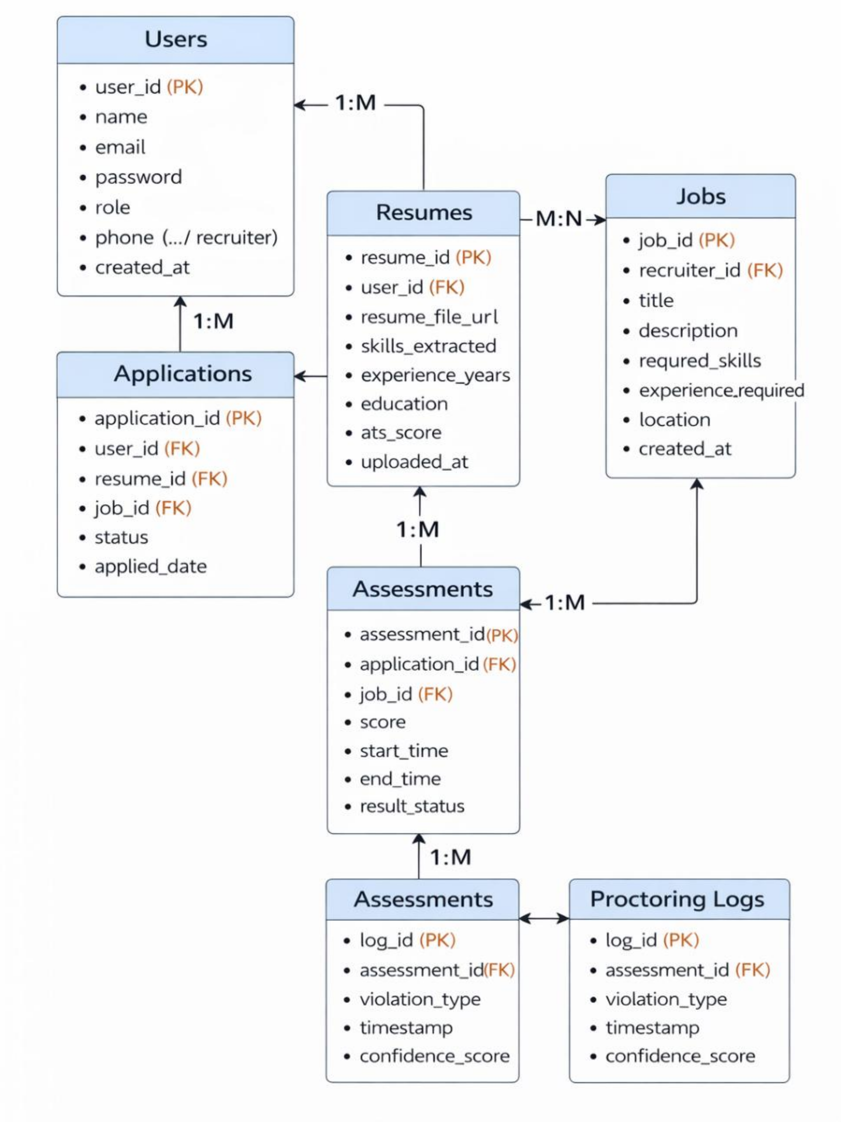



*Figure 5Database Design*

The database design supports secure storage and retrieval of recruitment data. A NoSQL database is used to ensure scalability and flexibility.

Database Collections

**1. Users**

Stores candidate and recruiter details.

- user_id
- name
- email
- password
- role (candidate/recruiter)
- profile_details

**2. Resumes**

Stores uploaded resumes and ATS scores.

- resume_id
- user_id
- resume_file_path
- extracted_skills
- ats_score
- upload_date

**3. Jobs**

Stores job postings created by recruiters.

- job_id
- recruiter_id
- job_title
- job_description
- required_skills
- experience_required
- posted_date

**4. Applications**

Tracks candidate applications.

- application_id
- user_id
- job_id
- application_status
- applied_date

**5. Assessments**

Stores test details and results.

- assessment_id
- job_id
- user_id
- score
- test_date

**6. Proctoring Logs**

Stores proctoring activity and violations.

- log_id
- assessment_id
- face_detected
- Object_detected
- multiple_faces
- tab_switch_count
- timestamp

# Chapter 7

# IMPLEMENTATION

## 7.1 Psuedocode

- **Ats Service**

BEGIN
INPUT:
Resume Text
Job Description
(Optional: Target Role Level, Industry Type, Preferred Skills List, Resume Format Type)

STEP 1: Preprocess Text
Convert resume text and job description text to lowercase.
Remove punctuation marks, special characters, and unnecessary whitespace.
Preserve numbers related to years of experience.
Tokenize both texts into individual words.
Generate bigrams and trigrams for phrase detection such as:
machine learning
data analysis
project management
Remove stop words such as:
the, is, at, which, on, and, for
Apply lemmatization to normalize words such as:
developing → develop
managed → manage
analyzing → analyze
Normalize technical synonyms such as:
ml → machine learning
ai → artificial intelligence
js → javascript
nlp → natural language processing
Return cleaned token lists for both resume and job description.

STEP 2: Identify Resume Sections
Detect structured resume sections including:
Professional Summary
Skills
Work Experience
Education
Projects
Certifications
Publications
Leadership
Contact Information
Store detected sections for later scoring evaluation.

STEP 3: Extract Keywords from Job Description
Extract important keywords using TF-IDF or noun phrase detection.
Classify extracted keywords into categories:
Technical Skills
Programming Languages
Tools and Frameworks
Cloud Platforms
Databases
Soft Skills
Certifications
Action Verbs
Store extracted job description keywords list.

STEP 4: Extract Keywords from Resume
Extract keywords from:
Skills section
Experience section
Projects section
Summary section
Certifications section
Classify extracted resume keywords into the same categories used for job description keywords.
Store extracted resume keyword list.

STEP 5: Extract Required Skills from Job Description
Identify skill importance levels using contextual phrases such as:
Required
Must-have
Mandatory
Preferred
Nice-to-have
Optional
Classify required skills into:
Must-have skills
Optional skills
Store required skills list.

STEP 6: Extract Skills from Resume
Scan the following resume sections:
Skills
Experience
Projects
Certifications
Create resume skills list.
Store extracted resume skills.

STEP 7: Calculate Keyword Match Score
Count number of exact keyword matches between job description and resume.
Count number of semantic keyword matches using synonym matching and phrase

similarity.
Apply scoring formula:
Keyword Match Score =
(Exact Matches × 1.0 + Semantic Matches × 0.75)
divided by total job description keywords
multiplied by 100
Add bonus points if keywords appear inside:
Professional Summary
Recent Experience Section
Project Titles
Cap keyword score at 100.

STEP 8: Calculate Skill Match Score
Compare resume skills with required skills.
Apply weighted scoring:
Must-have skill match weight = 2
Optional skill match weight = 1
Bonus skill match weight = 0.5
Calculate total matched weighted skills.
Apply scoring formula:
Skill Match Score =
Matched Skill Weight divided by Total Required Skill Weight multiplied by 100
Add bonus if skills appear in recent roles within last 3 years.
Cap skill score at 100.

STEP 9: Calculate Experience Score
Extract years of experience from resume.
Detect role progression indicators such as:
Intern
Junior
Engineer
Senior Engineer
Lead
Manager
Architect
Compare extracted experience with job role expectations:
Entry Level = 0–2 years
Mid Level = 2–5 years
Senior Level = 5+ years
Assign proportional score based on match.
Add bonus points if action verbs detected such as:
Developed
Managed
Architected
Optimized
Implemented
Led
Designed
Deployed

Scaled
Add leadership bonus if resume contains:
Team Leadership
Mentoring
Stakeholder Collaboration
Project Ownership
Cap experience score at 100.

STEP 10: Calculate Education Score
Detect highest degree level:
PhD
Master’s Degree
Bachelor’s Degree
Diploma
Assign base score:
PhD = 100
Master’s = 90
Bachelor’s = 80
Diploma = 65
Add bonus points if degree field matches job domain.
Add bonus points for relevant certifications such as:
AWS
Azure
Google Cloud
PMP
Scrum Master
Cisco
CompTIA
Machine Learning Certifications
Cap education score at 100.

STEP 11: Calculate Resume Format Score
Check presence of required sections:
Contact Information
Summary
Skills
Experience
Education
Projects
Assign format score based on section completeness.
Check formatting structure:
Bullet points used
Consistent spacing
Proper headings
Chronological order
Readable layout
Check ATS compatibility indicators:
No tables
No images

No complex graphics
Standard fonts used
Assign final format score.
Cap format score at 100.

STEP 12: Calculate Resume Length Score
Count total number of resume words.
Apply scoring logic:
Less than 300 words = Low Score
300 to 700 words = Optimal Score
700 to 1200 words = Acceptable Score
More than 1200 words = Reduced Score
Assign length score accordingly.
Cap length score at 100.

STEP 13: Calculate Section Presence Score (New Enhancement)
Check presence of advanced sections:
Projects
Certifications
Leadership Experience
Publications
Volunteer Work
Add bonus score if present.
Cap section presence score at 100.

STEP 14: Calculate Semantic Similarity Score (New Enhancement)
Use embedding similarity comparison between:
Resume content
Job description content
Compute cosine similarity score.
Normalize similarity score between 0 and 100.
Store semantic similarity score.

STEP 15: Compute Final Weighted ATS Score
Apply weighted scoring formula:
Final ATS Score =
(Keyword Match Score × 0.30) +
(Skill Match Score × 0.25) +
(Experience Score × 0.20) +
(Education Score × 0.10) +
(Format Score × 0.10) +
(Length Score × 0.05)
Add optional enhancements:
(Section Presence Score × 0.03)
(Semantic Similarity Score × 0.07)
Normalize final score to remain between 0 and 100.

STEP 16: Generate Detailed Report
Provide breakdown including:

Keyword Match Score
Skill Match Score
Experience Score
Education Score
Format Score
Length Score
Section Presence Score
Semantic Similarity Score
Highlight:
Matched Keywords
Missing Keywords
Matched Skills
Missing Skills
Provide improvement suggestions such as:
Add missing required skills
Improve keyword density
Include measurable achievements
Add certifications
Strengthen summary section
Improve formatting for ATS readability
Optimize resume length

OUTPUT:
Final ATS Score between 0 and 100
Detailed Score Breakdown Report
Personalized Resume Improvement Suggestions
END

- **Quiz Service**

```
BEGIN

FUNCTION generateQuiz(jobDescription, resumeText, role):
  TRY:
    Fetch questions from database
    IF database has questions:
      Select balanced set (3 easy, 4 medium, 2 hard, 1 analytical)
    ELSE:
      IF jobDescription AND resumeText:
        Generate adaptive quiz based on resume and job description
      ELSE:
        Generate role-specific quiz questions
    Shuffle selected questions
    RETURN shuffled quiz
  CATCH error:
    RETURN default role-specific quiz
FUNCTION getQuestionsFromDatabase():
  TRY:
```

```
        Fetch easy, medium, hard, analytical questions
        RETURN merged list
    CATCH:
        RETURN empty list
FUNCTION selectBalancedQuestionsFromDB(allQuestions):
    Separate questions by difficulty
    Select random sample:
        3 easy + 4 medium + 2 hard + 1 analytical
    RETURN combined list

    FUNCTION calculateQuizScore(answers):
        IF no answers:
            RETURN 0
        totalScore = 0
        maxScore = 0
        FOR each answer IN answers:
            points = difficultyPoints(answer.difficulty)
            Add points to maxScore
            IF answer.isCorrect:
                Add points to totalScore
        RETURN (totalScore / maxScore) * 100

    FUNCTION difficultyPoints(difficulty):
        Map difficulty levels to points:
            easy=1, medium=2, hard=3, analytical=4
        RETURN points

    FUNCTION detectCheating(proctoringLog):
        IF log empty:
            RETURN false
        Filter events from last 5 minutes
        Count violation types:
            no_face, multiple_faces, head_turned, tab_switch, window_minimize
        APPLY cheating rules:
            IF any rule exceeded threshold:
                RETURN true
        RETURN false

    FUNCTION validateQuizCompletion(candidate):
        IF quiz not assigned:
            RETURN invalid
        IF no answers submitted:
            RETURN invalid
        IF cheating detected (rejected):
            RETURN invalid
        RETURN valid
```

## 7.2 Screenshots

- **Home Page**

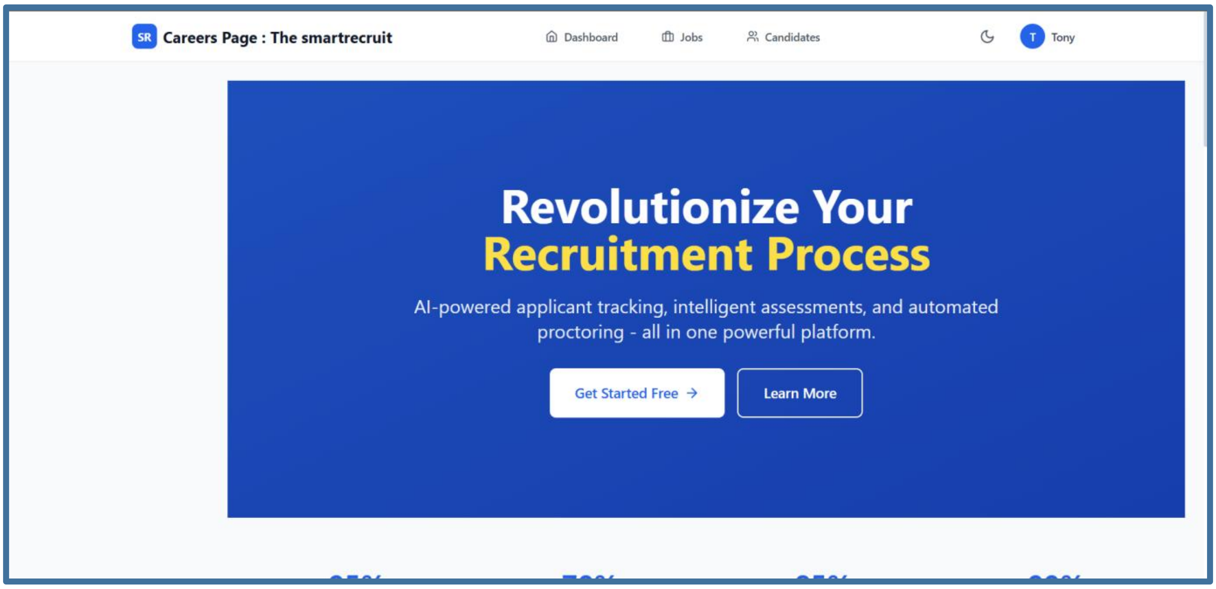



*Figure 7.1 Home Page*

- **Login Page**

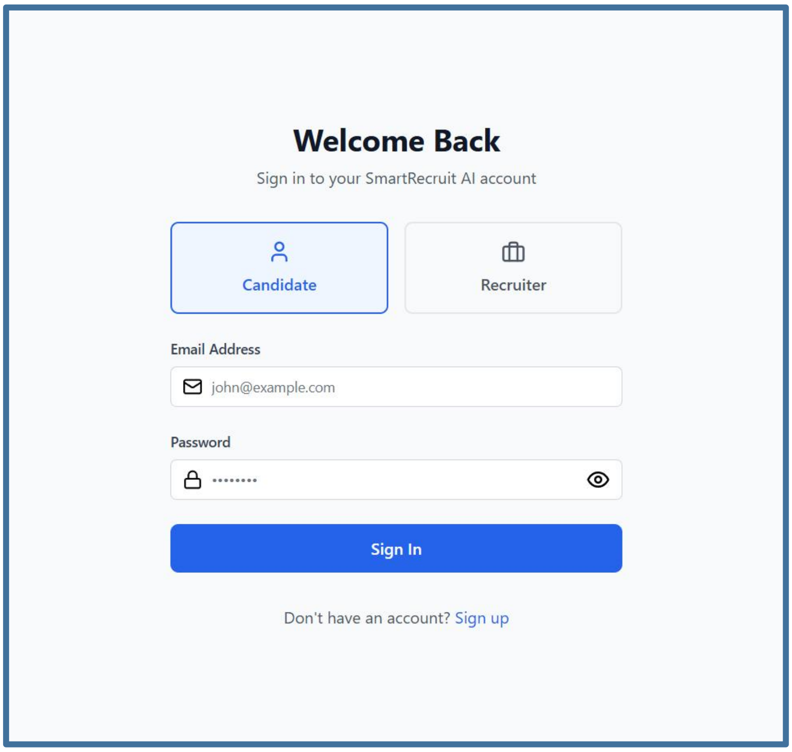



*Figure 7.2 Login Page with roles*

- **Candidate Dashboard**

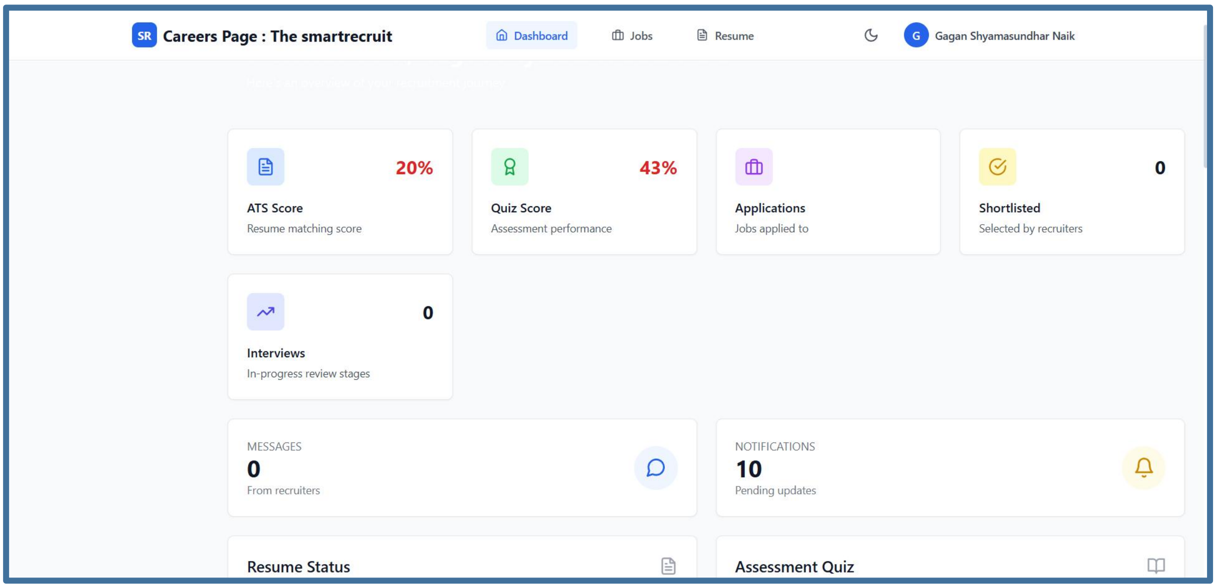



*Figure 7.3 Candidate Dashboard*

- **Ats Scoring : job role based**

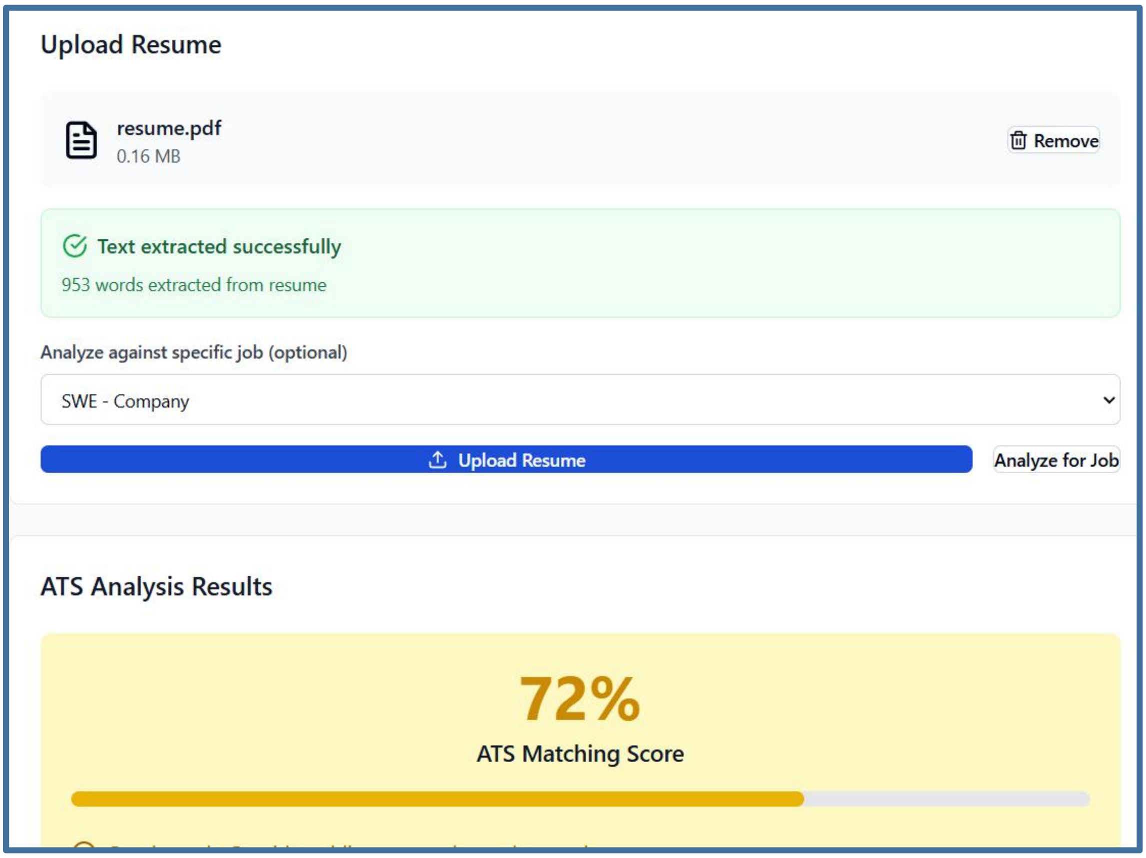



*Figure 7.4 Ats scoring*

- **Recruiter Dashboard**

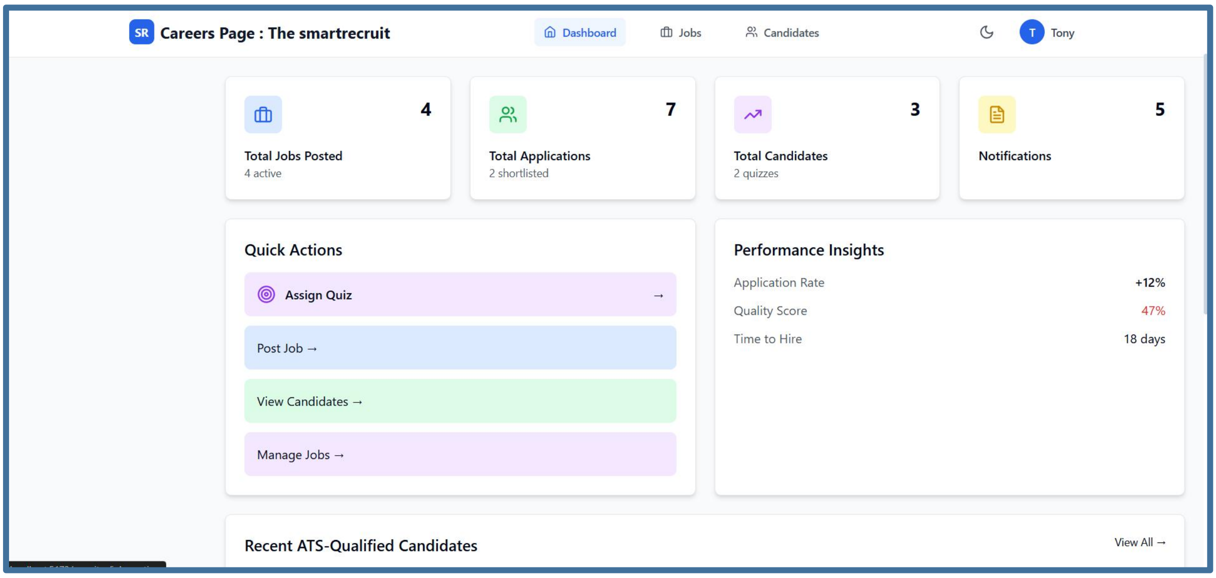
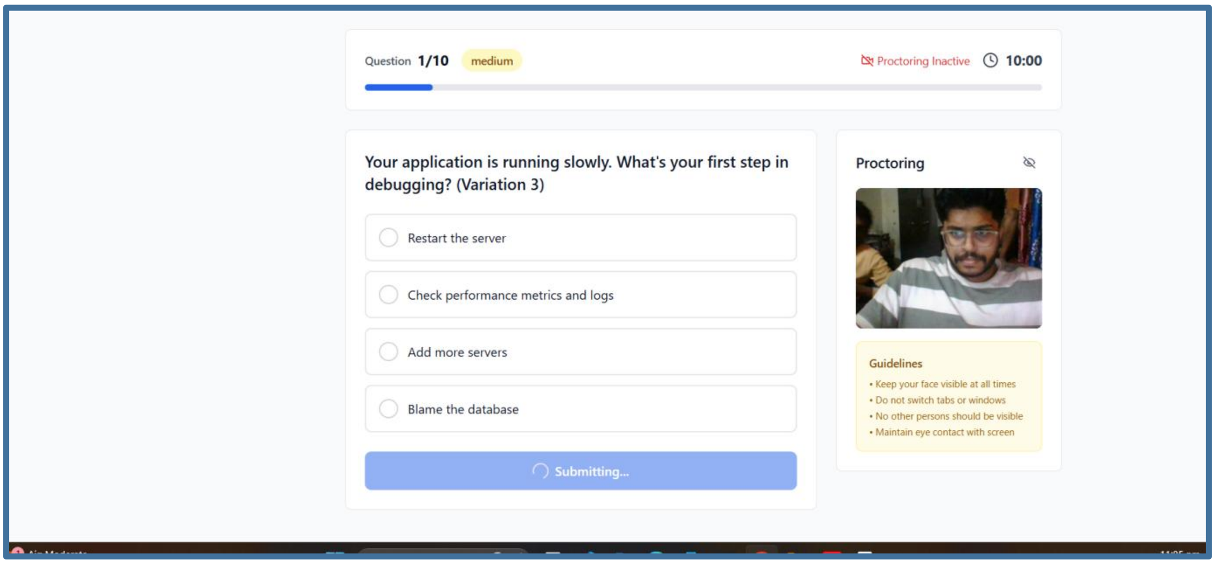



*Figure 7.5 Recruiter Dashboard*

- **Quiz**



# Chapter 8

# Software Testing

## 8.1 Manual Test Cases

- **Authentication and profile test cases**

| ID | Description | Input / Steps | Expected Result | Type | Status |
|---|---|---|---|---|---|
| A001 | Register with valid details | New email, password, role (candidate /recruiter) | Account created, redirected to d ashboard | Functional | Pass |
| A002 | Register with exist ing email | Already registered email | Error: “Email already exists” | Functional (Nega tive) | Pass |
| A003 | Login with valid cr edentials | Registered email & password | User logged in, dashboard loads | Functional | Pass |
| A004 | Login with invalid credentials | Incorrect email/password | Error: “Invalid credentials” | Functional (Nega tive) | Pass |
| A005 | Logout from syste m | Click logout button | User logged out successfully | Functional | Pass |
| A006 | Update profile wit h valid data | Skills, education, experience | Profile updated and saved | Functional | Pass |
| A007 | Update profile wit h invalid data | Empty required fields | Validation error messages show n | Functional (Nega tive) | Pass |

- **Resume upload and Ats**

| ID | Description | Input / Steps | Expected Result | Type | Status |
|---|---|---|---|---|---|
| B001 | Upload valid r esume | PDF/DOC resume | Resume uploaded success fully | Functional | Pass |
| B002 | Upload invalid file format | Image / unsupported file | Error: “Invalid file format ” | Functional (N egative) | Pass |
| B003 | ATS scoring g eneration | Resume + Job description | ATS score generated | Functional | Pass |
| B004 | Skill extractio n | Resume upload | Skills extracted correctly | Functional | Pass |
| B005 | View ATS fee dback | Click score details | Improvement suggestions displayed | Functional | Pass |

- **Job Search and Application**

| ID | Description | Input / Steps | Expected Result | Type | Status |
|---|---|---|---|---|---|
| C001 | View job listings | Open job search page | Jobs displayed | Functional | Pass |
| C002 | Apply for job | Click apply | Application submitted | Functional | Pass |
| C003 | Apply without login | Apply button clicked | Redirect to login page | Functional (Negative) | Pass |
| C004 | Track application status | Open application history | Status displayed correctly | Functional | Pass |

- **Quiz Service**

| ID | Description | Input / Steps | Expected Result | Type | status |
|---|---|---|---|---|---|
| Q001 | Start quiz successfully | Click Start Quiz | Quiz session begins and timer starts | Functional | Pass |
| Q002 | Load job-based questions | Start quiz for selected job | Questions loaded based on job skills | Functional | Pass |
| Q003 | Randomize question order | Refresh quiz session | Questions appear in random order | Functional | Pass |
| Q004 | Enforce time limit | Wait until timer ends | Quiz auto-submitted | Functional | Pass |
| Q005 | Submit quiz manually | Click Submit | Answers saved and quiz submitted | Functional | Pass |
| Q006 | Prevent multiple submissions | Click submit twice | Second submission blocked | Functional (Negative) | Pass |
| Q007 | Handle unanswered questions | Skip a question and submit | Unanswered marked incorrect | Functional | Pass |
| Q008 | Store quiz responses | Answer questions and submit | Responses saved in database | Functional | Pass |
| Q009 | Calculate quiz score | Submit quiz | Score calculated correctly | Functional | Pass |
| Q010 | Display quiz result | Open result page | Score displayed to candidate | Functional | Pass |
| Q011 | Log tab switching | Switch browser tab | Violation recorded | Functional | Pass |
| Q012 | Detect no face during quiz | Turn off camera | Warning shown & violation logged | Functional (Negative) | Pass |
| Q013 | Detect multiple faces | Another person appears | Cheating alert logged | Functional (Negative) | Pass |
| Q014 | Generate proctoring report | End quiz session | Proctoring report generated | Functional | Pass |
| Q015 | Link quiz result to recruiter view | Recruiter opens applicant | Quiz score visible to recruiter | Functional | Pass |

# Chapter 9

# Conclusion

The Careers Page: Smart Recruit system was conceptualised and developed to overcome the shortcomings of existing recruitment platforms with the goal of providing an intelligent, integrated and automated recruitment support platform. The platform effectively combines ATS-based resume similarity scoring, NLP-based skill extraction, role-specific online assessments, proctoring and recruiter feedback dashboards into an efficient and transparent recruitment process that enhances the hiring process.

By adopting similarity scoring algorithms like the Jaccard Similarity and Cosine Similarity, the system improves resume matching with job requirements beyond traditional keyword search-based methods. The use of secure MCQ and coding tests with real-time proctoring capabilities guarantees integrity in online candidate assessments, particularly in off-site recruitment settings. The recruiter dashboard also offers organised insights and comparative analysis metrics to accelerate and inform recruitment decisions.

The system shows how the integration of Artificial Intelligence, Natural Language Processing techniques, Computer Vision algorithms and full-stack web development (React.js, Node.js, Express.js and MongoDB) technologies can help automate manual tasks, eliminate biases in initial screening, and enhance candidate experience by providing clear hiring processes.

In summary, the Smart Recruit Careers Page module achieves its goal of providing an intelligent recruitment support system that boosts candidate-job fit, enhances assessment credibility, and optimises recruiter efficiency while ensuring human involvement in the final hiring process. The project demonstrates the usefulness of AI-powered recruitment systems in contemporary recruitment processes and lays the groundwork for future improvements in areas such as explainable AI scoring, predictive hiring analytics, and enterprise-wide deployment..

# Chapter 10

## Future Enhancement

While the Careers Page: Smart Recruit platform provides successful implementation of smart resume matching, assessment management and decision support for recruiting, there are a number of improvements that can be made to increase scalability, accuracy and ease of use in practical recruitment settings.

A key improvement would be the addition of predictive candidate ranking models using machine learning that can learn from past hiring decisions to suggest the best candidates for the job. This will enhance accuracy of decision-making and further ease the workload of recruiters.

A further enhancement would be the use of Explainable Artificial Intelligence (XAI) methods to explain ATS similarity scores. This would allow recruiters and candidates to understand how similarity scores were determined, and the factors that contributed to the score, allowing them to better understand the scoring results and increasing confidence in the system.

The system can also be enhanced to include AI-powered video interviews, applying computer vision and speech recognition methods to assess the candidate's confidence, communication, and other factors during the interview. This will supplement MCQ and coding tests.

The system can also be integrated with cloud deployment platforms like Amazon Web Services (AWS), Microsoft Azure or Google Cloud to further enhance the system for scalability, reliability and speed in large enterprise recruitment systems. It would enable the system to handle thousands of users.

A further improvement could be the addition of real-time recruiter recommendation dashboards using analytics. This would include predictive hiring, skill-demand predictions and workforce planning to guide recruitment strategies.

The system can further be enhanced by adding automated resume improvement feedback to candidates, where targeted feedback can help improve resumes of candidates by highlighting missing skills, syntax, and role-specific requirements.

Additionally, support for multi-language resume parsing and evaluation can make the platform suitable for global recruitment environments and diverse candidate pools.

Other potential enhancements to the system may include integration with professional job platforms like LinkedIn, Naukri and Indeed, to allow automated job sourcing and efficient recruitment processes through a unified platform.

Finally, support for mobile applications for both candidates and recruiters would enhance user experience and convenience by enabling mobile access and interaction with the recruitment system on-the-go.

In summary, such improvements will make the Smart Recruit platform more scalable, intelligent and enterprise-ready to support predictive analytics-based recruitment strategies and global sourcing needs.

# Appendix A

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# Appendix B

## PLAGARISM REPORT

# Appendix C

# POSTER